

Cluster 34

To view the latest Map of Science clusters and data, visit the live Map at sciencemap.eto.tech.

Subjects

Research fields: Computer science, Mathematics, Engineering

The most common research fields of articles in this cluster. [Read more.](#)

Research subfields: Data mining, Pattern recognition, Information retrieval

Common research subfields of articles in this cluster. [Read more.](#)

Key concepts: Content-Based Image Retrieval, image retrieval system, based image, retrieval system based, CBIR system

These key concepts are identified algorithmically using article titles and abstracts. [Read more.](#)

78.03% of all articles are AI-related

Articles are algorithmically labeled as AI-related or not. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.76% of English-language articles are related to robotics

Articles are labeled as robotics-related or not using an ETO model. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

68.18% of English-language articles are related to computer vision

Articles are labeled as computer vision-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to natural language processing

Articles are labeled as NLP-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to AI safety

Articles are labeled as related to AI safety or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.76% of English-language articles are related to cybersecurity

Vital signs

140 recent articles in the cluster

140 articles in this cluster were published in the five year period ending in 2023.

Average article is **17.81** years old

The average age of articles in this cluster. However, most Map of Science metrics are calculated using articles from the past five years only.

Growth rating: **0.19th** percentile

Over the three year period ending in 2023, this cluster grew faster than roughly 0.19% of other clusters.

Citation rating: **37.75th** percentile

A cluster's citation rating is equal to the average citation percentile for all the articles in the cluster. Citation percentiles are calculated relative to articles of the same publication year. In this case, the average article in this cluster is cited more often than roughly 37.75% of all other articles published in the same year.

93.57% of articles are in English

Not expected to grow extremely fast

As determined by ETO's growth prediction model. [Read more.](#)

Countries and languages

Top author countries

Each row lists the number of articles in this cluster from the last five years with at least one author from an organization in the specified country. Data available for 113 of 140 articles. 89 of these articles had at least one author from an organization in these ten countries.

Country	Articles (five year period ending in 2023)	Yearly citations per article (average)
India	50	0.63
China	17	0.18
Indonesia	8	0.44
Malaysia	4	1.18
United States	3	0.27
Brazil	3	0.56
United Kingdom	2	1.10
Philippines	2	0.75
South Korea	2	1.00
Bangladesh	2	0.50

Top funding countries

Data available for 5 of 140 articles. 5 of these articles disclosed funding from one or more organizations in these ten countries.

Country	Articles (last five years)	Yearly citations (average)
Brazil	2	0.83
South Korea	1	2.00
Taiwan	1	0.75

Articles and sources

Core articles

Core articles are especially highly connected to other papers within the cluster.

[The Electronic Waveform of Action Potentials in the Brain](#)

2020: Journal of Consciousness Exploration & Research. 1 citations.

[Retire away Essential Accuracy for Darkness Discovery and Elimination](#)

2020: TEST Engineering & Management. 1 citations.

2020: Baghdad Science Journal. 1 citations.

[A New CBIR Search Engine with a Vision Transformer Architecture](#)

2023: Lecture notes in networks and systems. 2 citations.

[A textual framework for contour retrieval using sub-multiple contour striking spread position learning](#)

2020: International journal of information technology. 4 citations.

[A comprehensive survey on the reduction of the semantic gap in content-based image retrieval](#)

2021: International Journal of Applied Pattern Recognition. 6 citations.

[Modified cuckoo search algorithm: Feature subset selection & Shape, Color and Texture Features Descriptors for Content-Based Image Retrieval](#)

2021: Journal of University of Shanghai for Science and Technology. 1 citations.

[Image retrieval system based on multi feature extraction and its performance assessment](#)

2021: International journal of information technology. 11 citations.

Review articles

Review articles have between 100 and 1000 out-citations. These articles describe and systematize other contributors' research.

[A retriever-independent framework for relevance feedback in content-based retrieval](#)

2021. 0 citations.

[Content-Based Image Retrieval: A Survey on Local and Global Features Selection, Extraction, Representation, and Evaluation Parameters](#)

2023: IEEE Access. 4 citations.

Highly cited articles

Highly cited articles are the articles in the cluster with the most citations overall.

[Textural and color descriptor fusion for efficient content-based image retrieval algorithm](#)

2020: Iran Journal of Computer Science. 23 citations.

[Low Cost Histogram Implementation for Image Processing using FPGA](#)

2020: IOP Conference Series: Materials Science and Engineering. 13 citations.

[A novel image retrieval technique based on semi supervised clustering](#)

2021: Multimedia tools and applications. 10 citations.

[‘Killer’ immune cells still recognize Omicron variant](#)

2022: Nature. 20 citations.

[A comprehensive survey on the reduction of the semantic gap in content-based image retrieval](#)

2021: International Journal of Applied Pattern Recognition. 6 citations.

[Region of interest-based image retrieval techniques: a review](#)

2020: IAES International Journal of Artificial Intelligence (IJ-AI). 19 citations.

[Local Tetra-Directional Pattern-A New Texture Descriptor for Content-Based Image Retrieval](#)

2020: Pattern Recognition and Image Analysis. (Advances in Mathematical Theory and Applications). 15 citations.

[Image retrieval system based on multi feature extraction and its performance assessment](#)

2021: International journal of information technology. 11 citations.

Top sources

The most common sources of articles from the last five years in this cluster, such as specific journals, conference proceedings, or preprint servers. Data available for 136 out of 140 articles. 26 of these articles came from these sources.

Source	Articles (last five years)
Lecture notes in networks and systems	5
International journal of information technology	4
Proceedings of SPIE	3
International Workshop on Artificial Intelligence and Cognition	2
Materials Today: Proceedings	2
International Journal for Research in Applied Science and Engineering Technology	2
Journal of Consciousness Exploration & Research	2
Smart innovation, systems and technologies	2
Multimedia tools and applications	2
IOP Conference Series: Materials Science and Engineering	2

Authors, organizations, and funders

Top authors

The authors with the most articles from the last five years in this cluster. Data available for 138 of 140 articles. 17 of these articles were written by these authors.

Author	Articles (last five years)	Yearly citations (average)
Shaheen Fatima	3	0.22
Ahmed Zeggari	2	0.25
Savita Sonoli	2	1.25
Fatima Shaheen	2	0.00
Priscila T. M. Saito	2	2.33
Fella Hachouf	2	0.25
Ramesh Kumar Sunkaria	2	5.10
Bradley Y. Bartholomew	2	1.20
Obed Appiah	2	0.75
G. Rosline Nesakumari	2	3.88

Top author organizations

The author organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 118 of 140 articles. 20 had at least one author from one of these ten organizations.

Organization	Articles (last five years)	Yearly citations (average)
Gulbarga University - India	3	0.11
Communication University of Zhejiang - China	3	0.07
Constantine 1 University - Algeria	2	0.25
Université Larbi Tébessi - Algeria	2	0.25
Punjab Technical University - India	2	0.58
Universidade Tecnológica Federal do Paraná - Brazil	2	0.83
Thapar Institute of Engineering & Technology - India	2	1
Abbès Laghrour University of Khenchela - Algeria	2	0.25
University of Energy and Natural Resources - Ghana	2	0.5
Jawaharlal Nehru Technological University, Hyderabad - India	2	0

Organization types

This table covers the 98 articles in this cluster from the five year period ending in 2023 with data about their author organization types (out of 140 articles total). [Read more](#)

Commercial	Education	Nonprofit	Government
1.48%	62.56%	1.48%	0.49%

Top industry organizations

The industry organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 3 out of 140 articles. 3 of these articles had at least one author from an organization listed below.

Organization	Articles (last five years)	Yearly citations (average)
Petroleum Geo-Services (United Kingdom)	1	0.8
Aditya Birla (India)	1	4
Dr. Reddy's Laboratories (India)	1	0

Top funders

Data available for 10 out of 140 recent articles. 6 of these articles disclosed funding from one or more of these organizations.

Funding organization	Articles (last five years)	Yearly citations (average)
National Council for Scientific and Technological Development - Brazil	2	0.83
Universidade Tecnológica Federal do Paraná - Brazil	2	0.83
Secretaria de Desenvolvimento Econômico Ciência Tecnologia e Ensino Superior - Brazil	2	0.83
Research and application of key technologies of multi video dynamic scene modeling for cultural heritage protection	2	0.10
Shanghai Key Lab of Modern Optical System	1	0.00
Healthy Aging Research Center, Chang Gung University from the Featured Areas Research Center Program of the Ministry of Education (MOE) in Taiwan	1	0.75
Fundação Araucária - Brazil	1	0.67
Ministry of Education - Brazil	1	1.00
City of Shanghai - China	1	0.00
PGS Exploration Ltd company	1	0.80

Funder types

This table covers the 5 articles in this cluster from the last five years with data about their funding organization types (out of 140 articles total).

Commercial	Education	Nonprofit	Government
0.00%	16.67%	8.33%	75.00%

Patents and industry

1218 patents cite articles in this cluster (**99.41th** percentile among all clusters)

7.47% of articles in the cluster are cited by patents

1.16% of articles have at least one author from industry

Top patent titles

Patent title	Same-cluster citations
AUTOMATED AND UNSUPERVISED GENERATION OF REAL-WORLD TRAINING DATA	9
VISUAL INTERACTIVE SEARCH	9
SELECTION OF INITIAL DOCUMENT COLLECTION FOR VISUAL INTERACTIVE SEARCH	8
LEXICON BASED SYSTEMS AND METHODS FOR INTELLIGENT MEDIA SEARCH	15
ADAPTIVE IMAGE RETRIEVAL DATABASE	14
AUTOMATIC GROUPING FOR USERS EXPERIENCING A SPECIFIC BROADCAST MEDIA	12
SYSTEM AND METHOD FOR TRACKING AND REWARDING MEDIA AND ENTERTAINMENT USAGE INCLUDING SUBSTANTITALLY REAL TIME REWARDS	12
SYSTEM AND METHOD FOR IMAGE ANNOTATION AND MULTI-MODAL IMAGE RETRIEVAL USING PROBABILISTIC SEMANTIC MODELS	12
HAND-DRAWN SKETCH RECOGNITION	11

Top patent assignees

Patent assignee	Same-cluster citations
MICROPORT (CHINA)	84
ALPHABET (UNITED STATES)	51
MICROSOFT TECHNOLOGY LICENSING, LLC	45
INTERNATIONAL BUSINESS MACHINES CORPORATION	39
IBM (UNITED STATES)	28
SALTLUX INC	24
PHILIPPINE RICE RES INSTITUTE PHILRICE	22
ADAPTTECH LTD	20
SHUTTERSTOCK INC	17

Connections

Top citing clusters

Cluster	Number of significant citations	CSET extracted phrases
29991	281	Content-Based Image Retrieval, image retrieval system, based image, image feature, CBIR system
1384	37	local binary pattern, Face Recognition, texture classification, LBP, Binary Pattern Based
10109	24	medical image retrieval, image retrieval system, content-based medical image, image features, convolutional neural network
6190	16	image retrieval, Deep convolutional features, Local CNN Features, Instance-level Image Retrieval, Convolutional Neural Network
2116	2	cross-modal hashing methods, Deep hashing, unsupervised cross-modal hashing, hash codes, Image Retrieval
4183	2	Shot Boundary detection, video shot boundary, boundary detection algorithm, gradual transition detection, based video shot
5094	1	shape recognition, shape features, shape matching, shape descriptor based, proposed method
40142	1	SAR Image Classification, SAR Images, remote sensing data, Earth Observation data, SAR data
29982	1	image retrieval, web image results, image search results, query image, click image search
1104	1	Similarity search, spatial data, metric spaces, spatial indexing structure, data sets

Top cited clusters

Cluster	Number of significant citations	CSET extracted phrases
29991	14	Content-Based Image Retrieval, image retrieval system, based image, image feature, CBIR system
1384	6	local binary pattern, Face Recognition, texture classification, LBP, Binary Pattern Based
10109	5	medical image retrieval, image retrieval system, content-based medical image, image features, convolutional neural network
4183	3	Shot Boundary detection, video shot boundary, boundary detection algorithm, gradual transition detection, based video shot
6190	3	image retrieval, Deep convolutional features, Local CNN Features, Instance-level Image Retrieval, Convolutional Neural Network
7694	3	Texture Features, Texture Image Classification, level co-occurrence matrix, Feature Extraction, Texture analysis methods
168	2	image feature matching, Feature Matching Algorithm, feature points, Matching Algorithm Based, ORB Algorithm
5580	2	information retrieval, Legal Document Retrieval, query expansion, Document retrieval methods, Legal Information Extraction
5269	1	Support Vector Machine, machine learning models, vector machine method, Vector Regression, SVM regression model
4084	1	manufacturing systems, Cyber-Physical Production Systems, Multi-agent system, Industrial Cyber-Physical Systems, Asset Administration Shell

Top GitHub repositories

Repository	Number of within-cluster citations	Number of stars
spotify/annoy	1	13218
pcpmartins/video-quality-assessment	1	52

The Map of Science runs on bibliometric data from [Clarivate Web of Science](#) and other sources. For details on our sourcing, methodology, and limitations, see the [tool documentation](#).